BUS 446
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Electric Vehicle Analysis

Data Analytics

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Data Set- Electric Vehicles

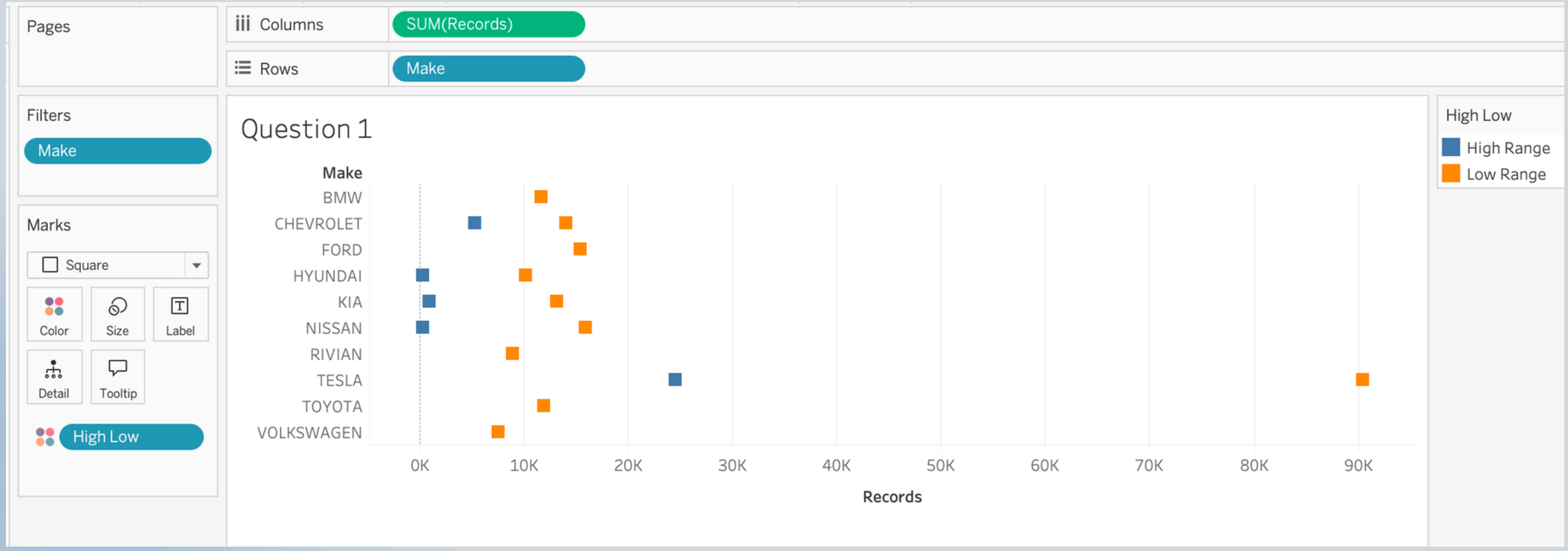
This dataset provides a comprehensive overview of the electric vehicle population, containing nearly 280,000 records of Battery Electric Vehicles and Plug-in Hybrid Electric Vehicles.

It captures detailed technical and geographic information for each vehicle, including the make, model, model year, and electric range, while also tracking Clean Alternative Fuel Vehicle eligibility and associated electric utilities.

Centered primarily on vehicles registered in Washington State, the data offers a granular look at EV adoption across different counties, cities, and legislative districts.

Question One:

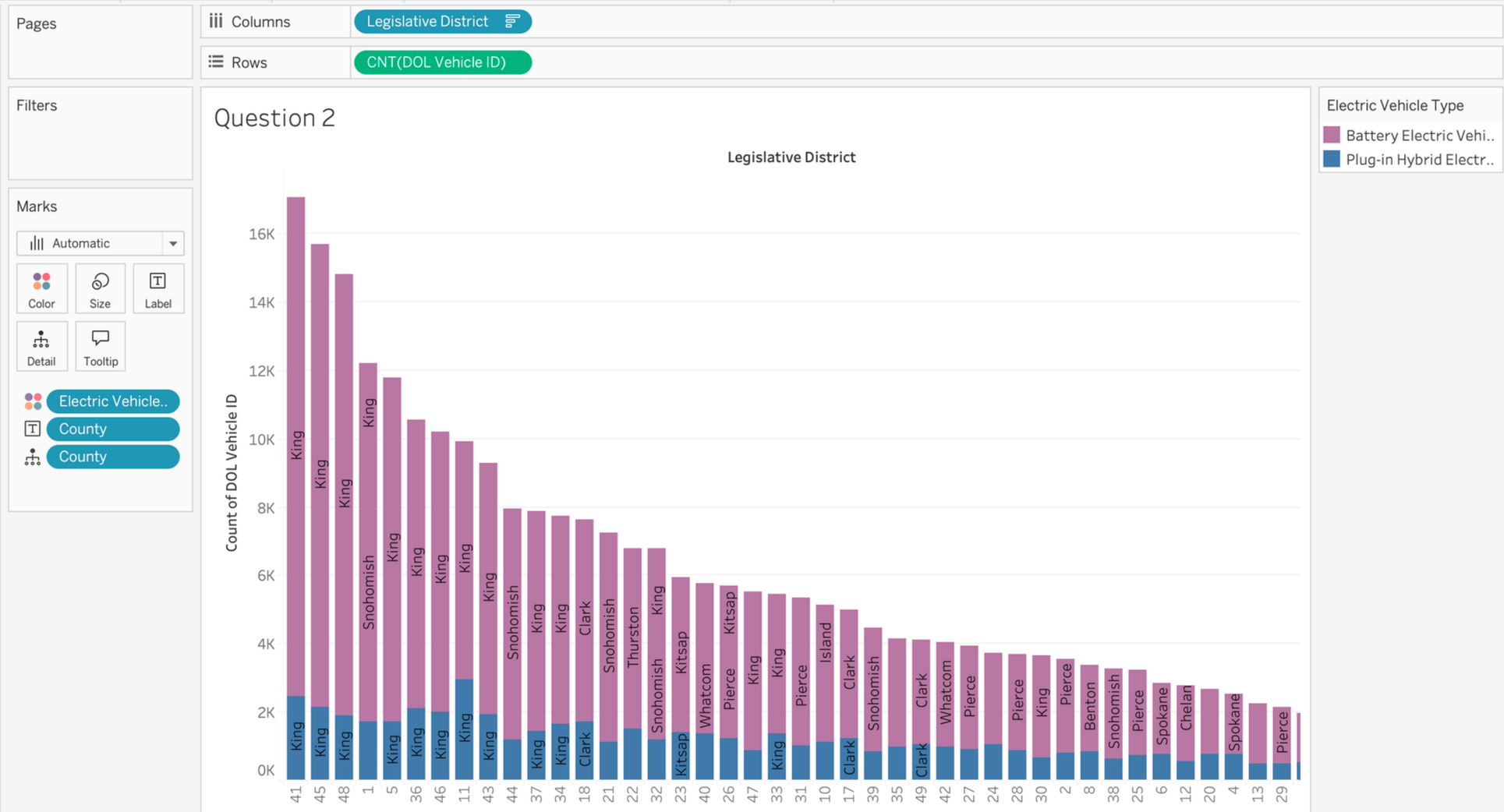
What are the most common vehicle makes and models in the dataset?



The provided visualization illustrates that Tesla is the overwhelmingly dominant manufacturer in the electric vehicle market, with its registrations far outpacing all other competitors combined. At the brand level, Tesla leads with over 114,000 vehicles, followed at a significant distance by Chevrolet and Nissan.

Question Two:

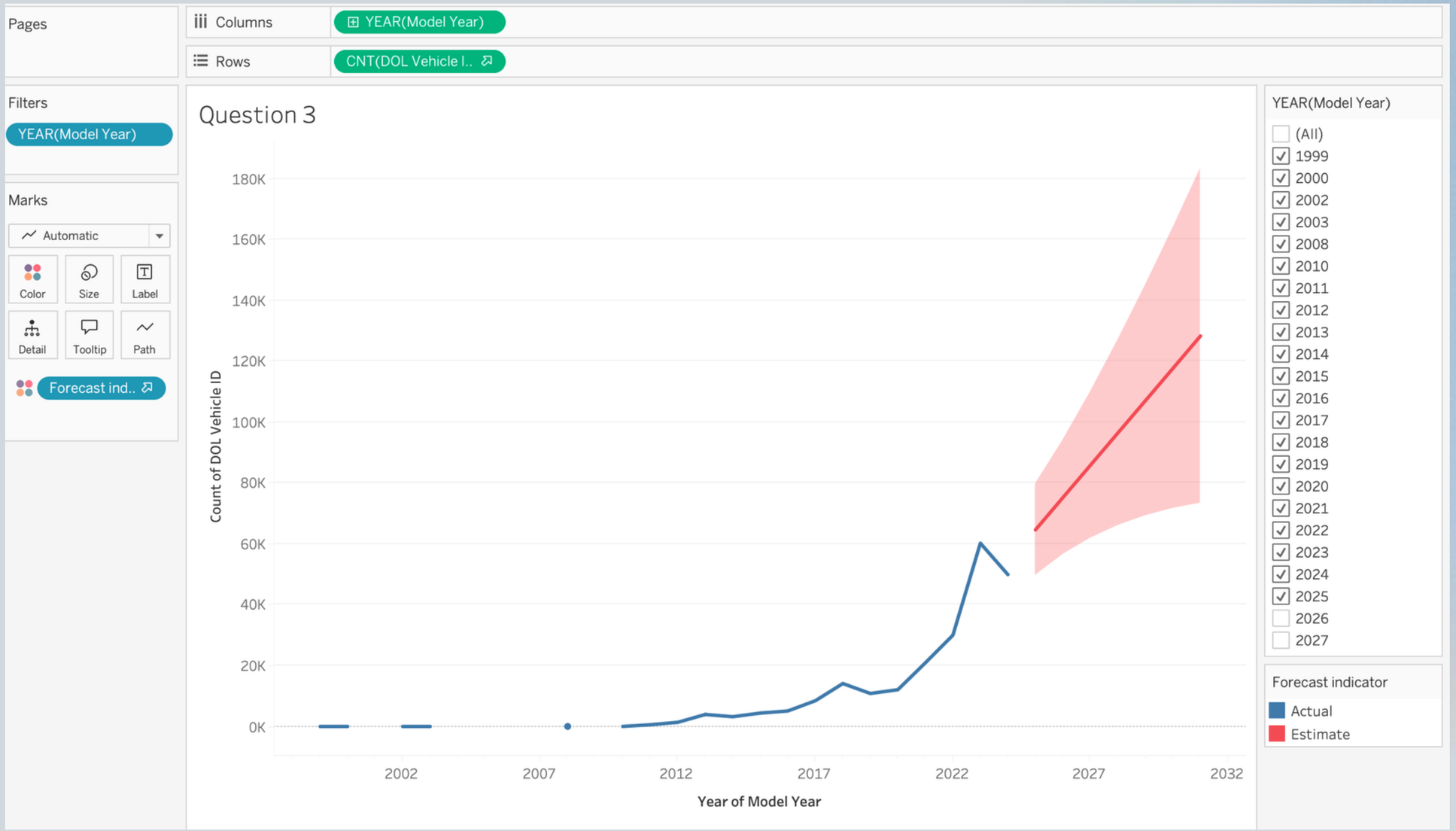
How does EV adoption vary across legislative districts?



Adoption peaks in District 41, King County, which leads with over 17,000 vehicles, followed closely by other "top-tier" districts within King County such as 45 and 48. These leading districts represent a significant portion of the total EV population, creating a "long tail" effect where the vast majority of districts have substantially lower registration numbers.

Question Three:

Can we forecast EV adoption trends using model year as a proxy?



Yes, model year serves as an effective proxy for forecasting EV adoption because it tracks the accelerating rate at which new electric vehicles are entering the fleet over time. The graph illustrates a distinct exponential growth pattern: adoption remained relatively flat for decades before beginning a steep, consistent climb around 2011. This upward trajectory represents a shift from early adopters to mass-market appeal.

How Was Tableau Helpful?

- Turn complex data into clear insights
- Highlight patterns and disparities
- Support decision-making
- Enable forecasting and trend analysis
- Makes data interactive and easier to explore

<u>Sorting & Filtering</u>

- Highlights dominant categories and filters allow users to isolate specific makes, models, or districts.

<u>Bar Charts</u>

- Helps quickly identify top performers
  - (Tesla, District 41)

<u>Visualization of Distribution</u>

- Tableau makes this pattern visually obvious, which would be hard to detect in a spreadsheet.

Thank You

Questions?